

IT Research Methodology

Conceptualizing a Research Design

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Prepared by: CHEN SOVANN

Outline

- What is a Research Design?
- The Functions of a Research Design
- Framework of Research Design
- Parameters of a Research Design
- Approaches of Research Design

❖ What is a Research Design

- A research design is a plan, structure and strategy of investigation so conceived as to obtain answers to research questions or problems
- The plan is the complete scheme or programme of the research.
- It includes an outline of what the investigator will do from writing the hypotheses and their operational implications to the final analysis of data. (Kerlinger 1986: 279)

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❖ What is a Research Design

- At the heart of the research lies in the understanding of the problem
- A researcher is capable to capture and identifying the need of the research and formulate the problem definition
- A research design is a procedural plan that is adopted by the researcher to answer questions **validly**, **objectively**, **accurately** and **economically**.

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❖ What is a Research Design

- A research design is the most important step in giving a direction to the research problem.
- It is the overall plan that deals with the aspects of complete design from the **study type**, **data collection approaches**, experimental designs, and **statistical approaches** for data samples.

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❖ Framework of Research Design

➢ A research framework deals with:

1. Defining precise, up to the point problem statement
2. Approaches/techniques that can be put to practice to collect samples/data
3. The details about the data that need to be analyzed and further researched upon
4. Approaches/experimental setups to be executed for the data processing and analysis.

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❖ Framework of Research Design

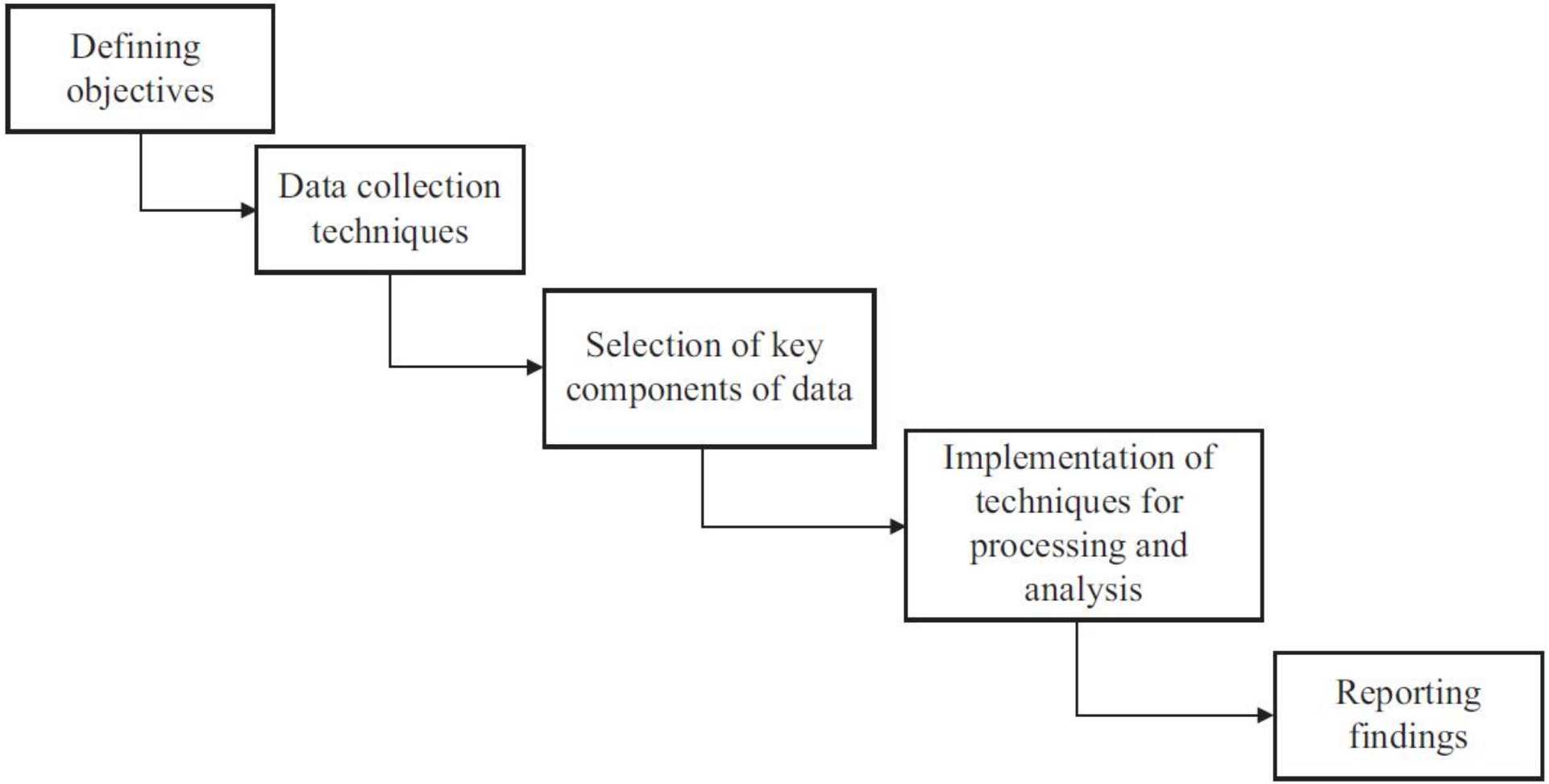


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❖ Parameters of a Research Design

➢ Based on the lines of framework, how the entire research would be placed is detailed as below:

(i) **Title of research**: This needs to be made clear by properly defining the title of the research. There should be no ambiguity; needs to be concise and clearly state what the researcher plans to do.

❖ Parameters of a Research Design

➢ Based on the lines of framework, how the entire research would be placed is detailed as below (Cont.):

(ii) **Importance of the research**: This section puts forth the significance of the work being carried out.

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❖ Parameters of a Research Design

➢ Based on the lines of framework, how the entire research would be placed is detailed as below (Cont.):

(iii) **Literature survey**: An understanding of which reference material needs to be applied and studied further on along with identification of shortfalls other researchers is important from the researcher's perspective. This aids in defining the scope of research and to have achievable objectives.

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❖ Parameters of a Research Design

➢ Based on the lines of framework, how the entire research would be placed is detailed as below (Cont.):

(iv) **Scope/objectives and problem statement**: A clear idea about the tasks that are required to be carried out as a part of research should be explicitly mentioned with the objectives. A problem statement defines the outcomes expected out of the research in a very precise way.

❖ Parameters of a Research Design

➢ Based on the lines of framework, how the entire research would be placed is detailed as below (Cont.):

(v) **Concepts of the terminology used**: A key aspect while coming up with research design is that the researcher needs to have understanding of the terms that would be using in the research.

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❖ Parameters of a Research Design

➢ Based on the lines of framework, how the entire research would be placed is detailed as below (Cont.):

(vi) **Data – Selection and collection and analysis**: Data collection is the most time-consuming task. Many times the researcher faces a problem of unavailability of the data. In this design phase, following factor should be taken care of from **where to get the data**, **how to collect**, **whether to use any specific equipment (form)**, **type of collection process**.

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❖ Parameters of a Research Design

➢ Based on the lines of framework, how the entire research would be placed is detailed as below (Cont.):

(vii) **Results: Interpretations and conclusions**: A very crucial aspect that a researcher should know is; how he plans to deliver the results and state his/ her claims while interpreting the results. Though a variety of methods – graphs, tables, scatter plots etc. can be used.

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❖ Parameters of a Research Design

➢ Based on the lines of framework, how the entire research would be placed is detailed as below (Cont.):

(viii) **Further research**: Once we begin with the planning of objectives to data collection, the analysis to results, the other aspect that one can specify is the further research.

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❖ Approaches of Research Design

- A measurable amount is referred as “quantity.”
- In case of qualitative research, its deals with subjective analysis. More of psychological aspects involved here.
- A survey about how much people are depressed if they spend time on social media can be an example of qualitative analysis.
- A very common example we can say about qualitative analysis is of social network sites.

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Q&A